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Surgeon Understanding and Perceptions of Social Determinants of Health



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ABSTRACT

Introduction: Social determinants of health impact surgical outcomes. Characterization of surgeon understanding of social determinants of health is necessary prior to implementation of interventions to address patient needs. The study objective was to explore understanding, perceived importance, and practices regarding social determinants of health among surgeons. **Methods:** Surgical residents and attending surgeons at a single academic medical center completed surveys regarding social determinants of health. We conducted semi-structured interviews to further explore understanding and perceived importance. A conceptual framework from the World Health Organization (WHO) Commission on Social Determinants of Health informed the thematic analysis.

Results: Survey response rate was 47.9% ($n = 69$, 44 residents [63.8%], 25 attendings [36.2%]). Respondents primarily reported good ($n = 29$, 42.0%) understanding of social determinants of health and perceived this understanding to be very important ($n = 42$, 60.9%). Documentation occurred seldom ($n = 35$, 50.7%), and referrals occurred seldom ($n = 26$, 37.7%) or never ($n = 20$, 29.0%). Residents reported a higher rate of prior training than attendings (95.5% versus 56.0%, $P < 0.001$). Ten interviews were conducted (six residents, four attendings). Residents demonstrated greater understanding of socioeconomic positions and hierarchies shaped by structural mechanisms than attendings. Both residents and attendings demonstrated understanding of intermediary determinants of health status and linked social determinants to impacting patients' health and well-being. Specific knowledge gaps were identified regarding underlying structural mechanisms including the social, economic, and political context that influence an individual's socioeconomic position. **Conclusions:** Self-reported understanding and importance of social determinants of health among surgeons were high. Interviews revealed gaps in understanding that may contribute to limited practices.

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Introduction

Social determinants of health describe contextual factors in a patient's life that function as predictors of access to and engagement in healthcare.^{1–3} The World Health Organization (WHO) established the Commission on Social Determinants of Health (CSDH) in 2005 to support partners in addressing social determinants of health-related factors to improve health and well-being.³ The Commission developed a conceptual framework that includes the social, economic, and political mechanisms that shape an individual's socioeconomic position (structural determinants), which lead to conditions of daily life that determine health status (intermediary determinants).³ Both the structural and intermediary determinants impact health and well-being.³ Among surgical patients, presence of social determinants of health-related needs have been associated with lower access to care and worse outcomes.^{4–7}

Knowledge and understanding of social determinants of health-related factors and their interrelated nature is important for clinicians in caring for patients. Ability to identify social determinants of health-related factors at the patient level may lead to improved risk stratification and perioperative optimization.^{8,9} Screening for social determinants of health-related factors has been trialed and successfully implemented in pediatric and primary care settings.^{10–12} However, widespread implementation may be hindered by healthcare providers lack of training and access to resources to address patient needs.^{13–15} Perceptions of social determinants of health have been described among primary care and hospital faculty in general.^{14–16} Perceptions of social determinants of health among surgical providers have not been explored.

Further insight into surgeon awareness and prioritization of social determinants health is necessary to enable development and implementation of interventions that better address social determinants of health-related needs among surgical patients. We sought to understand the current perceptions regarding social determinants of health among surgeons in our Department of Surgery (DOS). The first aim was to conduct a department-wide survey to evaluate surgeon understanding of, perceived importance of, and current clinical practices regarding social determinants of health. The second aim was to conduct in-depth semi-structured interviews to explore these concepts in greater detail and identify specific gaps in knowledge and understanding. We hypothesized that understanding and perceived importance would be relatively high, particularly among residents, and that current practices would be limited.

Methods

Setting and participants

We designed a convergent mixed-methods study and collected data through a survey and semi-structured interviews. All participants were part of the DOS of a single, tertiary academic medical center located in an urban setting

serving a diverse patient population. Participants included surgical residents (residents) and attending surgeons (attendings). The study was evaluated by the institutional review board (IRB) and determined to be exempt from full review.

Survey development and distribution

The research team designed a survey to evaluate current understanding of, perceived importance of, and clinical practices regarding social determinants of health ([Supplemental Digital Content 1](#)). The questions were informed by two validated surveys that explored knowledge of and practices regarding social determinants of health.^{13,17} The survey included a definition of social determinants of health on the first page: “the conditions in the environments where people are born, live, learn, work, play, worship, and age that affect a wide range of health, functioning, and quality-of-life outcomes and risks.” Clinical practices included screening patients for social determinant of health-related needs, capturing social determinants of health data in the electronic health record (EHR), and making referrals to resources that may address needs related to social determinants of health. Responses were recorded on a Likert scale from 1 to 5, with 5 representing the most positive response and 1 representing the most negative. Respondents were asked the proportion of their patients who had social or economic hardships that impact their medical care and compliance. The survey queried respondent demographics, including age, sex, years in practice, specialty, race, ethnicity, childhood economic class, and highest level of parental education. Respondents were also asked about any prior training related to social determinants of health. Two external, methodological experts reviewed the survey and provided targeted feedback. The platform SurveyMonkey (Momentive Inc San Mateo, CA, USA) was used for survey design and distribution.

All surveys began with a statement of consent. Participation was entirely voluntary with no incentivization. The survey was distributed and remained open from September 13, 2021 until October 11, 2021. Participants were recruited by email, WhatsApp (WhatsApp LLC, Menlo Park, CA, USA), and during department-wide lectures. Emails were distributed to the entire department three times with variation in day of the week and time to increase response rate. The survey link was distributed weekly through a WhatsApp group that included all residents. Finally, the survey QR code was displayed at the end of department-wide lectures such as grand rounds.

Interview guide and procedures

The semi-structured interview guide was developed by the research team prior to survey distribution thus not informed by survey responses ([Supplemental Digital Content 2](#)). Interview questions were designed through a collaborative team effort with multiple iterations until consensus was reached. Additional review was conducted by one external, methodological expert. This report focuses on findings related to 1) individual understanding of health disparities and social determinants of health, 2) perceived importance of DOS member understanding of these topics in caring for patients.

A convenience sampling approach was used as participants were recruited to be interviewed at the end of the survey. All those who expressed interest in being interviewed to further share their views were interviewed. Interviews were conducted between October 15, 2021 and December 8, 2021, prior to survey analysis. All interviews occurred at least 1 mo after the interviewee had taken the survey. A single interviewer (G.A.S.) with prior training in qualitative interviewing conducted all semi-structured interviews. The interviewer was personally known to all interviewees. Recognizing that this could potentially introduce acquiescence bias or social desirability bias, the interviewer attempted to frame questions in an open-ended fashion, and supported answers that seemed truthful with rephrasing of questions as necessary. All participants were read an informed consent statement and verbally consented. Interviews were conducted either via phone and recorded using Rev Call Recorder (Rev, San Francisco, CA, USA) or via Zoom (Zoom, San Jose, CA, USA) and recorded using the platform. The interviews lasted between 15 and 45 min. Interviews were transcribed verbatim using Trint (Trint Ltd London, UK) and deidentified prior to analysis.

Survey analysis

Continuous data were described using medians and interquartile range (IQR). Categorical data were described using counts and percentages. Demographic data were compared between residents and attendings using Wilcoxon rank-sum for continuous variables and Fisher's exact test for categorical. Likert scaled responses were treated as interval data and described as medians with IQR. Comparison between resident and attending responses were made using Wilcoxon rank-sum test. Analysis was done using RStudio (Version April 1, 2017, Rstudio PBC, Boston, MA, USA).

Qualitative interview analysis

Three researchers used inductive and deductive logic guided by the WHO CSDH Conceptual Framework to code transcripts and identify themes (Fig. 1).³ The CSDH Conceptual Framework illustrates the mechanisms that create social hierarchies (structural determinants) and the resulting determinants of health status (intermediary determinants) that impact health and well-being. The structural mechanisms include the social, economic, and political context that gives rise to socioeconomic positions stratified by income, education, occupation, gender, and race/ethnicity. The downstream conditions of daily life described by the term intermediary determinants include material circumstances, psychosocial circumstances, behavioral or biological factors, and the health system itself.

The analysis team (G.A.S., Y.G., Z.M.P) began by reading all transcripts. A representative interview was used to generate codes in a session with all three coders. Each transcript was then read and coded individually, followed by sessions of all coders meeting to reconcile differences in coding. The CSDH Conceptual Framework was used to inform development of themes related to individual understanding of social determinants of health. Quotes were categorized based upon the predominant theme recognizing that multiple elements of the framework may be identified within a single quote given the interrelated nature of these concepts. MAXQDA supported coding and indexing of the data (Standard 2020, VERBI Software, Berlin, Germany).

Results

Survey results

Of the 144 DOS members, 69 (47.9%) completed the survey including 44 residents (63.8%) and 25 attendings (36.2%)

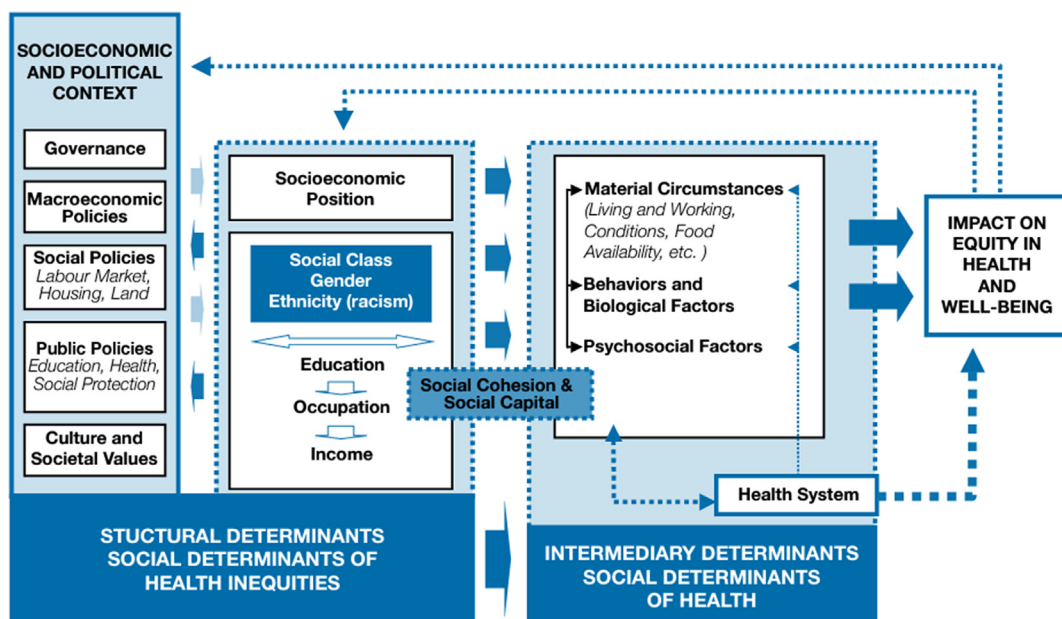


Fig. 1 – World health organization commission on social determinants of health conceptual framework. Reprinted with permission.

(Table 1). Race, ethnicity, childhood economic class, and highest level of parental education did not differ between residents and attendings. Of resident respondents, 39 (88.6%) were in the general surgery program, 2 (4.5%) in the plastic and reconstructive surgery program, and 3 (6.8%) in the urology program. Of attending respondents, surgical division were as follows: 2 (8.0%) abdominal transplantation, 2 (8.0%) colon and rectal surgery, 2 (8.0%) general surgery, 5 (20.0%) minimally invasive and bariatric surgery, 3 (12.0%) pediatric surgery, 5 (20.0%) plastic and reconstructive surgery, 3 (12.0%) surgical oncology, and 3 (12.0%) urology. A greater proportion of residents ($n = 42$, 95.5%) as compared to attendings ($n = 14$, 56.0%) reported prior training regarding social determinants of health ($P < 0.001$).

Respondents self-rated their understanding of social determinants of health as very good ($n = 12$, 17.4%) or good ($n = 29$, 42.0%) (Fig. 2). When asked how understanding social determinants of health was important in providing patient care, most responded very important ($n = 42$, 60.9%) or important ($n = 23$, 33.3%). Regarding practices related to social determinants of health, respondents reported that screening occurred often ($n = 22$, 31.9%) or sometimes ($n = 23$, 33.3%). Documentation in the EHR was primarily reported to occur seldom ($n = 35$, 50.7%). Referral to internal resources that may address patient needs was reported to occur sometimes ($n = 19$, 27.5%), seldom ($n = 20$, 29.0%) or never ($n = 17$, 24.6%). Referral to external resources were reported to occur seldom ($n = 26$, 37.7%) or never ($n = 20$, 30.0%). Respondents primarily estimated that 21%-40% ($n = 17$, 24.6%), 41%-60% ($n = 29$, 42.0%) or 61%-80% ($n = 18$, 26.1%) of patients were faced with social or economic hardships that impact their medical care and compliance. There were no differences in self-rated understanding, perceived importance, or clinical practices between residents and attendings (Supplementary Table S1).

Qualitative interview results

Of the 69 survey respondents, 10 (14.5%) participated in an interview. Interviewees included six residents (60.0%) and four attendings (40.0%), four of which were female (40.0%) and six male (60.0%). For the attendings, years in practice ranged from 1 y to greater than 20 y. Residents were represented from all post graduate year levels.

Surgical residents demonstrate understanding of socioeconomic positions and hierarchies shaped by structural mechanisms

Representative quotes that indicated DOS member understanding of structural determinants are included in Table 2. While concepts related to individual socioeconomic position and stratification by income education, occupation, gender, and race/ethnicity were explored by residents, these stratifiers were not directly discussed by attendings. No DOS members mentioned the larger social, economic, and political context.

When asked to define social determinates of health, one resident stated that social determinants of health were “socioeconomic factors that can influence one’s access to health care and their outcomes with their health issues, things like obviously finance, social standing, education level as it pertains to health literacy, social support.” In this resident’s definition, key stratifiers of socioeconomic position including

education, income, social class (social standing) were described together with inclusion of social support, an intermediary determinant. Inclusion of intermediary determinants with structural determinants was common, suggesting an awareness of the interrelated nature of these concepts. Further, concepts of class, education, occupation, and income, were often described together. These factors are closely related as income and education can be understood as outcomes of the social stratification processes, while occupation can even be considered a proxy for social stratification.³

Inclusion of race, ethnicity, and gender in defining social determinants of health was less frequent. One resident linked the recognition of these factors with provider bias and the impact on outcomes:

“...their age, their ability, their sex, their race, their ethnicity, I think that can bias providers. And sometimes we can forget that patients are coming to us from a variety of backgrounds, particularly in terms of their historical experience with the health care system and the way we treat them, interact with them. I think they care and I hope outcomes are better when we are cognitive of the experience they’ve had before.”

Another resident alluded to accounting for a patient’s ethnic background by considering language without directly discussing race or ethnicity.

Both surgical residents and attending surgeons demonstrate understanding of intermediary determinants of health status
Representative quotes that indicated DOS member understanding of intermediary determinants are included in Table 3. In defining and discussing social determinants of health, all DOS members identified various elements related to intermediary determinants.

Of the three elements related to socioeconomic position (material circumstances, behaviors and biological factors, and psychosocial factors), material circumstances were the most commonly described by interviewees. Many referenced appropriate housing conditions and ability to afford food or medications through financial means. Psychosocial factors were also frequently discussed. Examples included lack of social support or stress caused by family situations. Behavioral and biological factors were the least referenced and most often related to issues regarding diet.

The healthcare system as an independent, intermediary determinant was often referenced in the context of access to care. Several interviewees also identified specific interventions that would address patient needs related to access. For example, one attending shared, “In our bowel management practice, we’ve had a lot of patients just not be able to come to appointments because they can’t get a ride. So we’re working on trying to create vouchers.” In this statement, the interviewee clearly defines a relationship between the patient need, transportation, and the lack of current system in place.

Surgical residents and attending surgeons link social determinants to impacting patients’ health and well-being

Table 4 contains representative quotes from interviewees that demonstrate understanding of the relationship of the

Table 1 – Characteristics of survey respondents from the department of surgery (DOS) and comparison between surgical residents and attending surgeons.

Characteristic	Total	Residents	Attendings	P-value
Total	69	44 (63.8)	25 (36.2)	
Age	31 (28-40)	29 (28-31)	44 (39.5-52)	<0.001
Sex				0.02
Female	27 (39.1)	22 (50.0)	5 (20.0)	
Male	39 (56.5)	20 (45.5)	19 (76.0)	
Prefer not to answer	3 (4.3)	2 (4.5)	1 (4.0)	
Years in practice	4 (2, 12.5)	2 (1-4)	17 (11-25)	<0.001
Race				0.30
Asian	12 (17.4)	10 (22.7)	2 (8.0)	
Black	1 (1.4)	1 (2.3)	0	
Caucasian	51 (73.9)	30 (68.2)	21 (84.0)	
Native American	2 (2.9)	1 (2.3)	1 (4.0)	
Prefer not to answer	3 (4.3)	2 (4.5)	1 (4.0)	
Ethnicity				0.99
Hispanic	5 (7.2)	3 (6.8)	2 (8.0)	
Non-Hispanic	61 (88.4)	39 (88.6)	22 (88.0)	
Prefer not to answer	3 (4.3)	2 (4.5)	1 (4.0)	
Economic class				0.27
Wealthy	8 (11.6)	3 (6.8)	5 (20.0)	
Upper middle class	16 (23.2)	12 (27.3)	4 (16.0)	
Middle class	32 (46.4)	20 (45.5)	12 (48.0)	
Lower middle class	6 (8.7)	3 (6.8)	3 (12.0)	
Poor	2 (2.9)	2 (4.5)	0	
Prefer not to answer	5 (7.2)	4 (9.1)	1 (4.0)	
Parent education				0.58
Did not finish high school	0	0	0	
High school diploma or equivalent	8 (11.6)	5 (11.4)	3 (12.0)	
Two-y college	0	0	0	
Four-y college	22 (31.9)	16 (36.4)	6 (24.0)	
Postcollege degree	36 (52.2)	21 (47.7)	15 (60.0)	
Prefer not to answer	3 (4.3)	2 (4.5)	1 (4.0)	
Prior training	56 (81.2)	42 (95.5)	14 (56.0)	<0.001

structural and intermediary determinants to an individual's health and well-being. All of the DOS members linked structural and/or intermediary determinants to clinical outcomes. However, many of the references were nonspecific in that they did not provide concrete examples of outcomes. The only specific example of an outcome was readmission, mentioned by one surgical resident. Discussion of well-being in the more general sense was also limited and included references to sleep and mental health.

Several residents discussed adjusting clinical decision making in consideration of structural or intermediary determinants. For example, "people at [institution name] will live there for months because of social stuff, they can't get a wheelchair, they can't get crutches, or we can't do home wound vac[uum]s...we don't take that stuff into account. Maybe we shouldn't be doing free flaps on people. Maybe we should be doing more amputations if that means they're going

to be in the hospital for 6 mo." The concept of adjusting clinical management in consideration of social determinants to improve outcomes was unique among the residents.

Discussion

An understanding of social determinants of health and potential implications for clinical outcomes is a necessary foundation to the delivery of equitable surgical care. We found that while self-rated understanding and importance of social determinants of health was high, in actual practice, screening, documentation, and referrals were seldom used. Specific knowledge gaps were identified using interviews. No members of the DOS discussed the social, economic and political structural mechanisms that shape an individual's socioeconomic position. Residents often mentioned stratifiers of

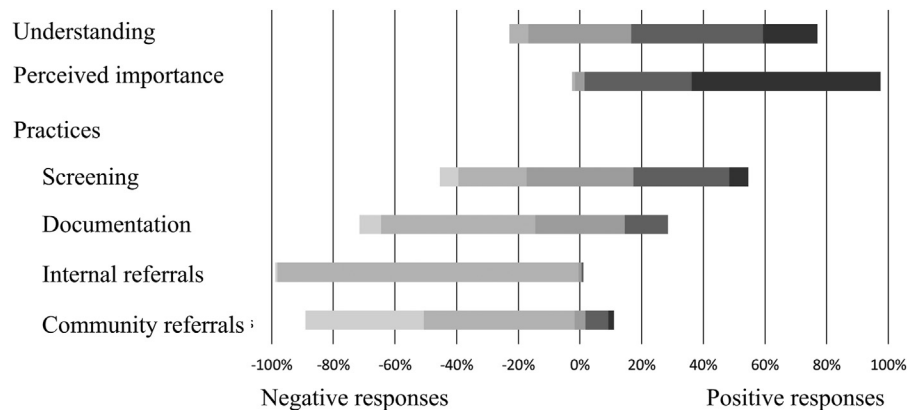


Fig. 2 – Surgeon understanding of, perceived importance of, and practices regarding social determinants of health (n = 69). Image represents the results from Likert Scale survey responses. Responses had five categories represented in the image; the darkest colors align with positive responses and the lightest negative. Neutral responses were centered on 0.

socioeconomic position such as income, education, occupation, gender, and race/ethnicity, while attendings were more focused on the intermediary determinants of health status including material circumstances, behavioral and biological factors, and psychosocial factors. All interviewees discussed the impact on equity in health and well-being, but in general terms without identifying linkages to specific outcomes.

Engaging in practices targeted toward mitigating social determinants of health-related problems for our patients can be challenging. Kostelanetz *et al.* 2022 found that among 193 healthcare professionals at an academic medical center, 93% recognized the importance of social determinants of health information in healthcare delivery, but less than half of participants commonly assessed these topics among patients.¹⁴ Barriers identified by clinicians to screen for social needs included lack of resources, time to ask, training, and staff support.^{14,15} These prior evaluations were within primary care or internal medicine specialties where screening for social needs may be perceived as more appropriate. Our findings showed a slightly greater gap between perceived importance and action than had previously been identified in the literature, which may be explained by the differences in specialty.

Several notable gaps in knowledge and understanding that were not elucidated as part of the surveys became clear from the interviews. The WHO CSDH Framework highlights the importance of the underlying social, political, and economic context and the structural mechanisms that “interplay between context and socioeconomic position: generating and reinforcing class divisions that define individual socioeconomic position within hierarchies of power, prestige, and access to resources.”³ Ultimately, these are the root cause of inequities in health. It may be that these underlying factors were not mentioned because they feel out of surgeons’ control to address. The intermediary determinants, or the downstream factors related to conditions of daily life, may be more apparent and tangible. This offers one explanation as to why the attendings, who had lower rates of formal training, more readily discussed the intermediary determinants. However, recognition of the root causes of inequities in health is

necessary in developing and supporting initiatives that work toward achieving health equity.

The information gathered from this study is being used to inform efforts at our institution targeted toward better identifying and addressing social determinants of health-related needs among surgical patients. Given that training around social determinants of health is being increasingly incorporated into medical school and residency curriculums, including at our institution, most residents graduate with some formal exposure.¹⁸ Understanding differences in resident and attending knowledge gaps has allowed us to target education appropriately. We have also been working toward implementing a screening program for social determinants of health-related needs in our outpatient surgical clinics. While some social determinants of health-related problems may be outside the ability of healthcare systems to address, programs particularly in housing, income support, and care coordination have been successful.^{19–21} For example, a spine clinic that implemented a screening program was able to identify five patients with social determinants of health-related needs and address these prior to surgery.⁸ Incorporation of screening tools into the EHR may also facilitate uptake of International Classification of Diseases, Tenth Revision, Clinical Modification Z codes (ICD-10-CM Z55-65).¹⁷ These nonbillable codes were included in ICD-10-CM to identify patients with social determinants of health-related needs and have been associated with worse outcomes.^{22,23} While Z codes cannot directly be billed for, the codes do add to patient complexity for outpatient visits. Through including individual social determinants of health-related needs in the EHR, we could potentially optimize management to improve clinical outcomes.⁹ Institutions may consider using the findings from this study to inform education initiatives at their own institutions, incorporate screening programs with actionable interventions, and increase documentation of social determinants of health-related needs in the EMR.

This study has limitations. First, we included surgeons only within one healthcare system where likely a shared institution-specific mentality and culture exists. Second, our

Table 2 – Representative quotes that demonstrate surgical provider understanding of the socioeconomic context, positions, and resulting hierarchies (structural determinants).

Framework element(s)	Quote
Education Income Occupation Ethnicity/racism Psychosocial (intermediary)	“Do they have health literacy to understand what surgery they’re having, to understand the alternatives, risk benefits? Can they afford the medications that they need?...Do they have health insurance to cover? Do they have a ride home? Do they have somebody to take care of them? Do they speak English?...understanding where different people are coming from, and knowing the different barriers that they can encounter, and being proactive and solving them or finding solutions to them...in the preop phase, I think it’s something that surgeons sometimes leave behind” – Surgical resident
Occupation Material (intermediary)	“We do a lot of hand surgery, and people can’t get their hand dressings wet or use their hands. A lot of the people who we take care of are carpenters or laborers or housekeepers... we just sort of willy nilly say, Oh, you need to take a month off of work’...and typically people can’t do that.” – Surgical resident
Education	“When you’re giving discharge instructions to your patients or explaining their medications to them, they might be nodding their head and saying, ‘Yeah,’ but I can tell you from personal experience with family members who have called me to ask me questions after a doctor’s appointment. They often don’t know what we’re talking about... I think that health literacy is a very overlooked part of social determinants of health that is upstream of a lot of these issues.” – Surgical resident
Gender Ethnicity/racism	“The way people present to us...whether it be their age, their ability, their sex, their race, their ethnicity, can bias providers. And sometimes we can forget that patients are coming to us from a variety of backgrounds, particularly in terms of their historical experience with the health care system and the way we treat them, interact with them. I think their care, and I hope outcomes are better when we are cognitive of the experience they’ve had before...often black patients who have had negative experiences in the past come in with a level of distrust that can impede care. And to get through that, I think we have to have an understanding of what their experience has been before” – Surgical resident

When present, intermediary determinants were also included. This demonstrates the interrelated nature of the framework elements.

results may be biased by convenience sampling, the proportion of resident respondents, and median age. For both surveys and interviews, convenience sampling was used, which

introduces motivation bias into the study. Motivation bias can be due to interest in the topic, a disgruntled point of view, or desire to support one’s own specific agenda.²⁴ The response

Table 3 – Representative quotes that demonstrate surgical provider understanding of determinants of health status (intermediary determinants).

Framework element(s)	Quote
Psychosocial Behaviors/biological	“If we look in the area of psychosocial conditions...that can be mental health concerns, anxiety and depression, it can be compliance. It can be substance abuse...we almost consider it to be prehabilitation or optimization of psychosocial issues prior to the procedure in order to enhance outcomes.” – Attending surgeon
Psychosocial Material	“I think there’s a lot of issues with making it to appointments and things for people who have three jobs and one car and family and multiple kids and things like that and don’t have the same help or money to pay for help or nannies or any of those lines. That does alter care too.” – Attending surgeon
Psychosocial Material Income (structural)	“I think about where people live and what their home situation is, who’s living with them at home, how many people they’re taking care of, who’s taking care of them and food insecurity. And how they’re paying for food, like whether they’re on Snap or EBT benefits and stuff like that, and basic housing structure. All those kind of things come to mind.” – Surgical resident
Behaviors/biological Health system Education (structural)	“The first thing that comes to mind is access to healthcare and not only access, but also quality the level of quality...as a bariatric surgeon, access and social determinants of healthy options, food eating options, healthy diet, exercise, you know, school education, the type of things all goes into kind of the primary prevention bucket.” – Attending surgeon
Material	“Social determinants of health are the things outside of the immediate health care system that still impact patients health, so living in a house with mold or lead, not having transportation to access adequate food or transportation to get health care...” – Surgical resident

When present, structural determinants that were discussed within the quote are also included. This demonstrates the interrelated nature of the framework elements.

Table 4 – Representative quotes that demonstrate surgical provider understanding of the impact of structural and intermediate determinants on equity in health and well-being.

Framework element(s)	Quote
Health system (intermediary)	"I think that making clinicians aware of these differences, how do you determine these differences and then how to appropriately react to these differences will reduce some of these barriers to care... outcomes are going to vary based on these limitations that they have and teaching clinicians how to recognize those and act appropriately should help maximize everybody's care." – Attending surgeon
Income (structural) Occupation (structural) Health system (intermediary)	"I think that there's a lot of things that we may not be aware of, and if we don't know to ask the right questions, the patient's outcomes can be significantly impacted. So for example, if we prescribe a medication and don't know that the patient doesn't have insurance or can't afford it, they may not be likely to fill their prescription. And then, depending on what it's for, may come back with worsening of their disease or have an adverse outcome...other things that I think it significantly affects are the patient's ability to obtain home health services, the ability to make it to follow up appointments. And you know, in the provider perspective, we might think of someone as being quote unquote noncompliant if we don't realize that that there are things that significantly impact the reasons patients make certain decisions." – Surgical resident
Health system (intermediary)	"There are certain times that when we're making a clinical decision, we will change based on a patient's social situation...you're doing what's safest and most durable for a patient based on their actual likelihood of being able to see you again based on what their health care access is. I think it's appropriate to change the care that you're going to get people because you have to do what's best for the patient and their situation." – Surgical resident

rate among residents was higher than attendings (47% versus 36%). Furthermore, the median age of attending respondents was 44. In consideration of these factors, our findings are likely biased toward an increased perceived understanding and importance. This would be due to motivation bias from interest in the topic and the increasing incorporation of topics related into social determinants of health into medical school curriculums.¹⁸ Third, we were unable to delineate experiences among surgical specialties given the sample size. We suspect that there would be differences in practices based upon the acuity of common diseases encountered and patient characteristics. Finally, our institution has a well-resourced case management team in place to support patients with social determinants of health-related needs. This may not be the case for all hospitals, and there are ethical considerations regarding screening in situations where resources do not exist to help address these needs for patients.²⁵

Conclusions

Surgeon self-reported understanding of social determinants of health was high; however, specific knowledge gaps were identified during interviews. Furthermore, current practices regarding identifying and addressing social determinant of health-related needs among surgical patients were limited. These findings may inform initiatives related to provider education and patient screening that work toward improving equitable delivery of surgical care and clinical outcomes.

Supplementary Materials

Supplementary data related to this article can be found at <https://doi.org/10.1016/j.jss.2023.08.050>.

Author Contributions

study conceptualization and design: GAS, ANS; data collection: GAS, YG, ZMP; analysis and interpretation of the results: GAS, YG, ZMP, JKJ, BCG, ANS; draft of manuscript: GAS, YG, ZMP, AD, MR; critical revision of manuscript: BCG, JKJ, ANS.

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Disclosure

None declared.

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